

Math 241, F04
Quiz 7

Name:
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (12 points) Given a function $f(x, y) = x^3 + y^3 - 6xy$,
a) Find its all critical points.

Solution: Let $\nabla f = 0$, then we have

$$f_x = 3x^2 - 6y = 0 \quad (1)$$

$$f_y = 3y^2 - 6x = 0 \quad (2)$$

From (1), we get $y = x^2/2$ and substitute it in (2), then obtain

$$3\left(\frac{x^2}{2}\right)^2 - 6x = 0$$

$$x^4 - 8x = 0$$

$$x(x^3 - 8) = 0$$

Thus, $x = 0$ or $x = 2$. Clearly $y = 0$ when $x = 0$ and $y = 2$ when $x = 2$. Therefore, the critical points are $(0, 0)$ and $(2, 2)$.

b) Determine whether each such a point gives a local maximum or minimum, or whether it is a saddle point using the second partials test.

Solution: $f_{xx} = 6x$, $f_{yy} = 6y$, and $f_{xy} = -6$, so

$$\begin{aligned} D = D(x, y) &= f_{xx}f_{yy} - f_{xy}^2 \\ &= (6x)(6y) - (-6)^2 \\ &= 36xy - 36 \end{aligned}$$

For the critical point $(0, 0)$, $D(0, 0) = -36 < 0$, so $(0, 0)$ is a saddle point.

For the critical point $(2, 2)$, $D(2, 2) = 36 \cdot 2 \cdot 2 - 36 = 108 > 0$ and $f_{xx}(2, 2) = 12 > 0$, so $f(2, 2) = 2^2 + 2^2 - 6 \cdot 2 \cdot 2 = -8$ is a local minimum.

Problem 2. (8 points) Find the minimum of $f(x, y) = x^2 + 4xy + y^2$ subject to the constraint $x - y - 6 = 0$.

Solution: First we have

$$\nabla f(x, y) = \langle 2x + 4y, 4x + 2y \rangle .$$

Let $g = x - y - 6 = 0$, then

$$\nabla g(x, y) = \langle 1, -1 \rangle .$$

The Lagrange equation becomes

$$2x + 4y = \lambda \quad (1)$$

$$4x + 2y = -\lambda \quad (2)$$

$$x - y - 6 = 0 \quad (3)$$

Let (1)+(2), we could find

$$6x + 6y = \lambda - \lambda$$

$$x + y = 0$$

$$y = -x$$

Substitute it in (3), we get

$$x - (-x) - 6 = 0$$

$$2x = 6$$

$$x = 3$$

Thus, $y = -3$. Therefore the minimum value of f subject to the constraint is $f(3, -3) = 3^2 + 4 \cdot 3 \cdot (-3) + 3^3 = -18$.